

Chapter 2 Notes

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1. **Theorem 2.20.** Let E be a set and $p \in E'$. Then every neighborhood of p contains infinitely many points of E .
2. **Corollary.** A finite point set has no limit points.
3. **Theorem 2.30.** Let X and Y be sets, $X \subseteq Y$, and $B \subseteq Y$ be open. Then $A \subseteq X$ is open relative to X if and only if $A = X \cap B$.
4. **Definition 2.31.** Let X be a metric space, $E \subseteq X$, and $\{U_i\} \subseteq X$ be a collection of open sets. Then $\{U_i\}$ is said to be an *open cover* of E provided $E \subseteq \bigcup_i U_i$.
5. **Definition 2.32.** Let X be a metric space and $K \subseteq X$. Then K is said to be *compact* provided every open cover of K contains a finite subcover. That is, if U_i is an open cover of K , then
$$\exists i_1, \dots, i_n \quad K \subseteq U_{i_1} \cup \dots \cup U_{i_n}.$$
6. **Theorem 2.33.** Let K , X , and Y be sets and $K \subseteq X \subseteq Y$. Then K is compact relative to X if and only if K is compact relative to Y .
7. **Theorem 2.34.** Let X be a metric space and $K \subseteq X$ be compact. Then K is closed.
8. **Theorem 2.35.** Let K be a compact set and $C \subseteq K$ be closed. Then C is compact.
9. **Corollary.** Let C be a closed set and K be a compact set. Then $C \cap K$ is compact.
10. **Theorem 2.36.** Let X be a metric space and $\{K_i\} \subseteq X$ be a collection of compact sets such that $\bigcap_{i=1}^n K_i \neq \emptyset$. Then $\bigcap_i K_i \neq \emptyset$.
11. **Corollary.** Let $\{K_n\}$ be a collection of nonempty compact sets such that $K_n \supseteq K_{n+1}$. Then $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.
12. **Theorem 2.37.** Let K be a compact set, $E \subseteq K$, and $|E| = \infty$. Then $E' \subseteq K$. That is, E has a limit point in K .

13. **Theorem 2.38.** Let $\{I_n\} \subseteq \mathbb{R}$ such that $I_n \supseteq I_{n+1}$. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.
14. **Theorem 2.41.** Let $E \subseteq \mathbb{R}^2$. Then the following are equivalent:
- (a) E is closed and bounded.
 - (b) E is compact.
 - (c) Every infinite subset of E has a limit point in E .
15. **Theorem 2.42.** Let $E \subseteq \mathbb{R}^n$ be bounded and $|E| = \infty$. Then $E' \subseteq \mathbb{R}^n$. That is, E has a limit point in $\mathbb{R}^n$.
16. **Definition 2.45.** Let X be a metric space and $A, B \subseteq X$. Then A and B are said to be *separated* provided $A \cap \text{Cl}(B) = \emptyset$ and $\text{Cl}(A) \cap B = \emptyset$.
17. **Definition 2.47.** $E \subseteq \mathbb{R}$ is connected if and only if
- $$\forall x, y \in E \quad x < z < y \implies z \in E.$$
18. **Exercise 2.8.** Let $E \subseteq \mathbb{R}^2$ be an open set and $x \in E$. Then $x \in E'$.